

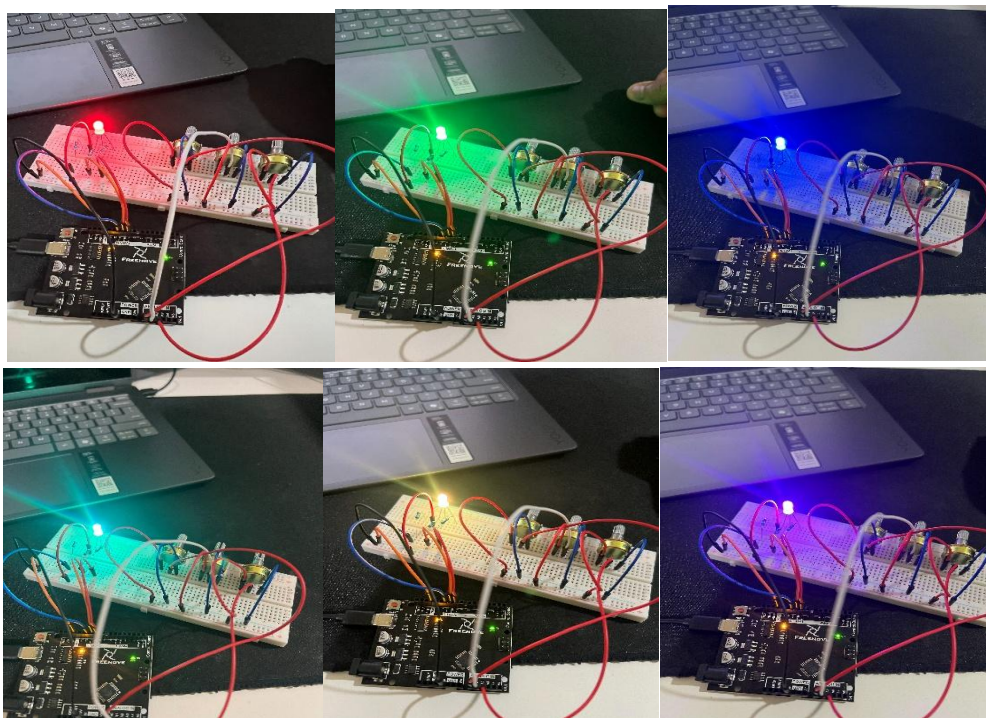
RGB LED Colour Mixer

Brief:

Implemented an interactive RGB LED colour mixer using an Arduino Uno, RGB LED, three rotary potentiometers, and three current-limiting resistors. Each potentiometer independently controls the brightness of one colour channel, red, green, or blue, allowing any colour to be created by hand. RGB LEDs mix colours and shades e.g. red + green produces yellow, red + blue produces magenta, green + blue produces cyan, and all three at full brightness produce white. The PWM duty cycle controls perceived brightness of each primary colour, so the entire visible colour spectrum can be created by adjusting the three potentiometers (except black).

Circuit Design

Each colour channel follows the same topology: a rotary potentiometer on an analog input pin provides a voltage between 0 V and 5V, which the ADC reads. A 220 Ω resistor sits in series between the Arduino PWM output pin and the corresponding LED anode. The Arduino output is 5V and a typical LED forward voltage is ~ 2 V, with a safe operating current of 15–20 mA. Applying Ohm's law: $R = (V_{cc} - V_f) / I = (5 - 2) / 0.015 \approx 200 \Omega$. 220 Ω is the nearest standard value above 200 Ω , limiting current to ~ 13 –14 mA. The sketch reads each potentiometer on ADC channels A0–A2 and maps the 10-bit result to a PWM duty cycle written to the corresponding LED pin. The map() function handles the scaling from the ADC range (0–1023) to the PWM range (0–255).



Video Demo:

<https://youtube.com/shorts/9eYDLcujW00?feature=share>